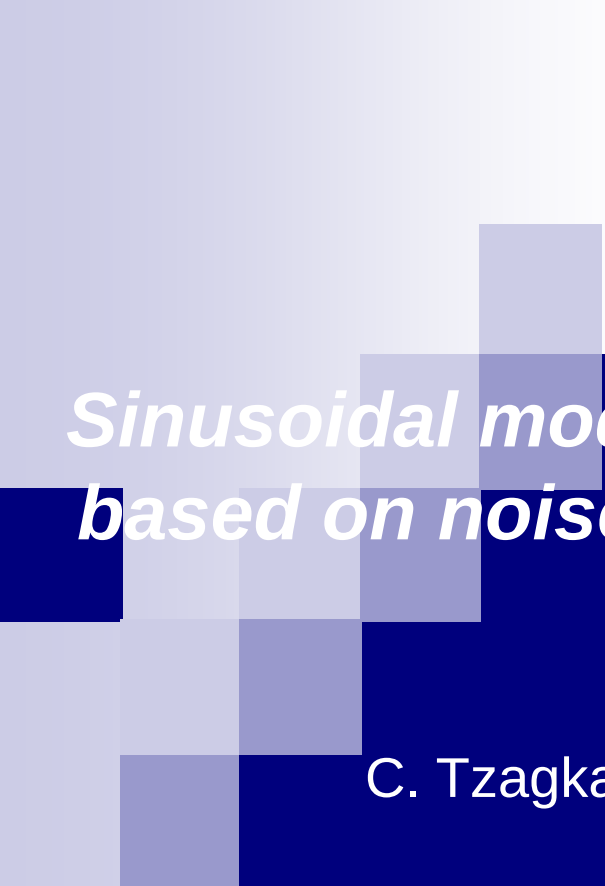




Sinusoidal modeling of spot microphone signals based on noise transplantation for multichannel audio coding

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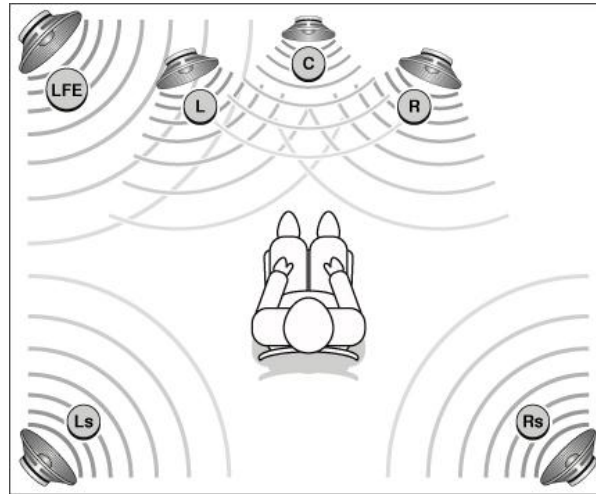


Outline

- Introduction
- Background
 - Information for multichannel recordings
 - Sinusoidal plus noise model
- Noise transplantation method
- Experimental results
- Conclusions

Introduction (1/3)

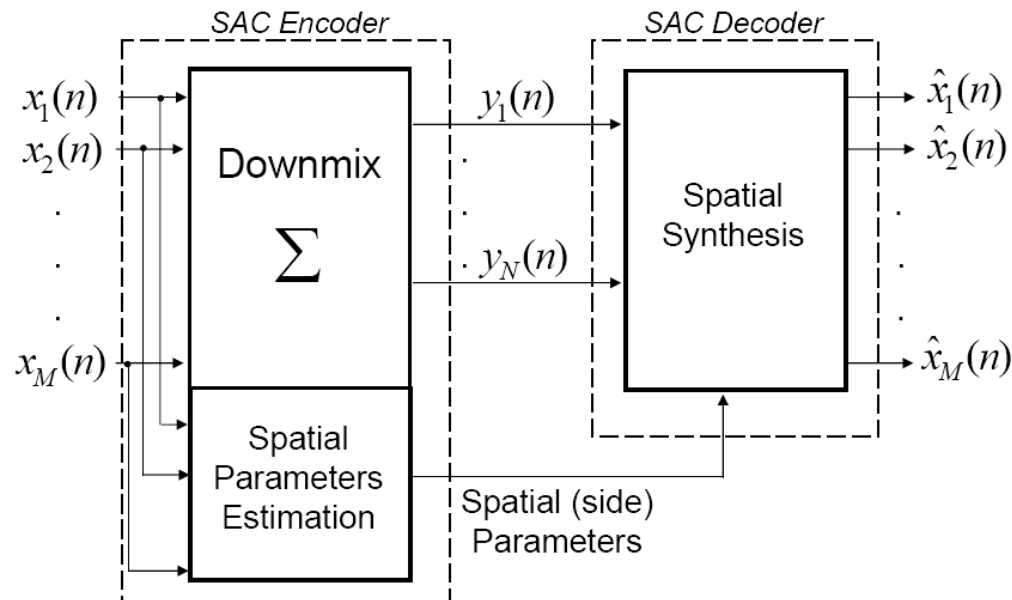
- Multichannel audio immerses the listener with a realistic acoustic scene



- Increased demand on high datarates because of the large number of channels
- Exploit inter-channel redundancy to reduce coding rate
- Towards this direction ***Spatial audio coding (SAC)*** was recently proposed

Introduction (2/3)

- The objective in **SAC** is to capture the spatial image of a multichannel audio signal



- **SAC** is an extremely low bitrate efficient representation method because of the reduced number of channels to be transmitted

Introduction (3/3)

- Produce a multichannel recording by mixing a large number of microphones
- Transmit the multiple microphones before the mixing process
 - ***remote mixing***
 - ***remote collaborations*** of geographically distributed musicians
- Introduce the sinusoidal plus noise ***transplantation*** approach for modeling the multiple signals ***before*** the mixing
 - focus on ***spot*** microphone signals' modeling
- Our method follows the **SAC** philosophy
 - model each spot signal with respect to a reference audio signal

Background Information (1/2)

Microphone signals of a multichannel recording

- A number of microphones (***stem recordings***) capture the characteristics of the recording venue
- Stem recordings are categorized into
 - ***Spot*** microphone signals: close to the sound source
 - ***Reverberant*** microphone signals: far from the sound source
- Spot signals recreate the sense that the sound source is distributed such as in an orchestra
- Reverberant signals capture the reverberation information of the venue

Background Information (2/2)

Sinusoidal plus noise model

- Audio signal is modeled as: $s(n) = \sum_{l=1}^L \underbrace{A_l(n) \cos(\theta_l(n))}_{\text{deterministic component}} + e(n)$
 - L : number of sinusoids
 - $A_l(n)$: instantaneous amplitude
 - $\theta_l(n)$: instantaneous phase
 - $e(n)$: modeling error (noise component)

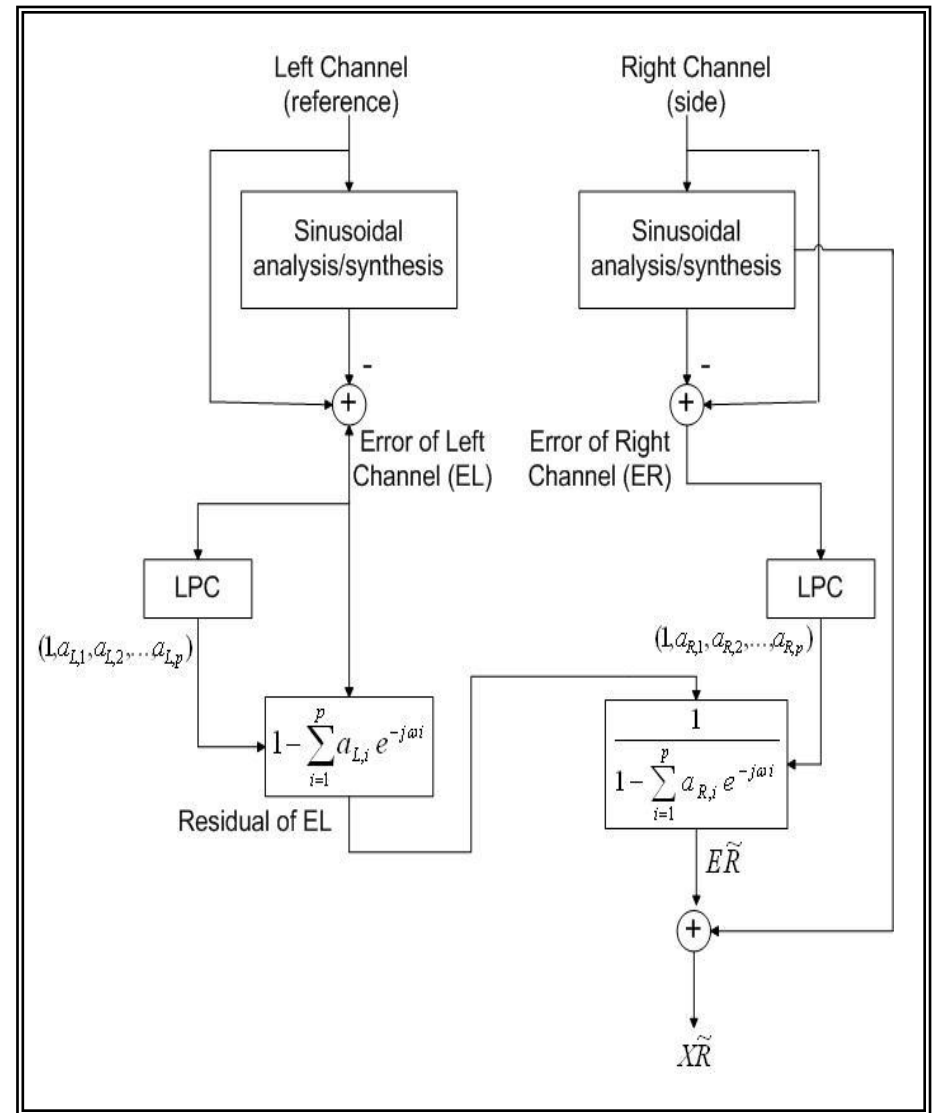
- Use Linear Prediction (LP) analysis to model the noise spectral envelope

$$e(n) = \sum_{i=1}^p a(i)e(n-i) + r_e(n) \quad \xRightarrow{\text{frequency domain}} \quad S_e(e^{j\omega}) = \left| \frac{1}{A(e^{j\omega})} \right|^2 S_{r_e}(e^{j\omega})$$

- $r_e(n)$: residual of the noise $e(n)$
- p : AR filter order
- $p+1^{\text{th}}$ -dimension vector α represents the spectral envelope of $e(n)$

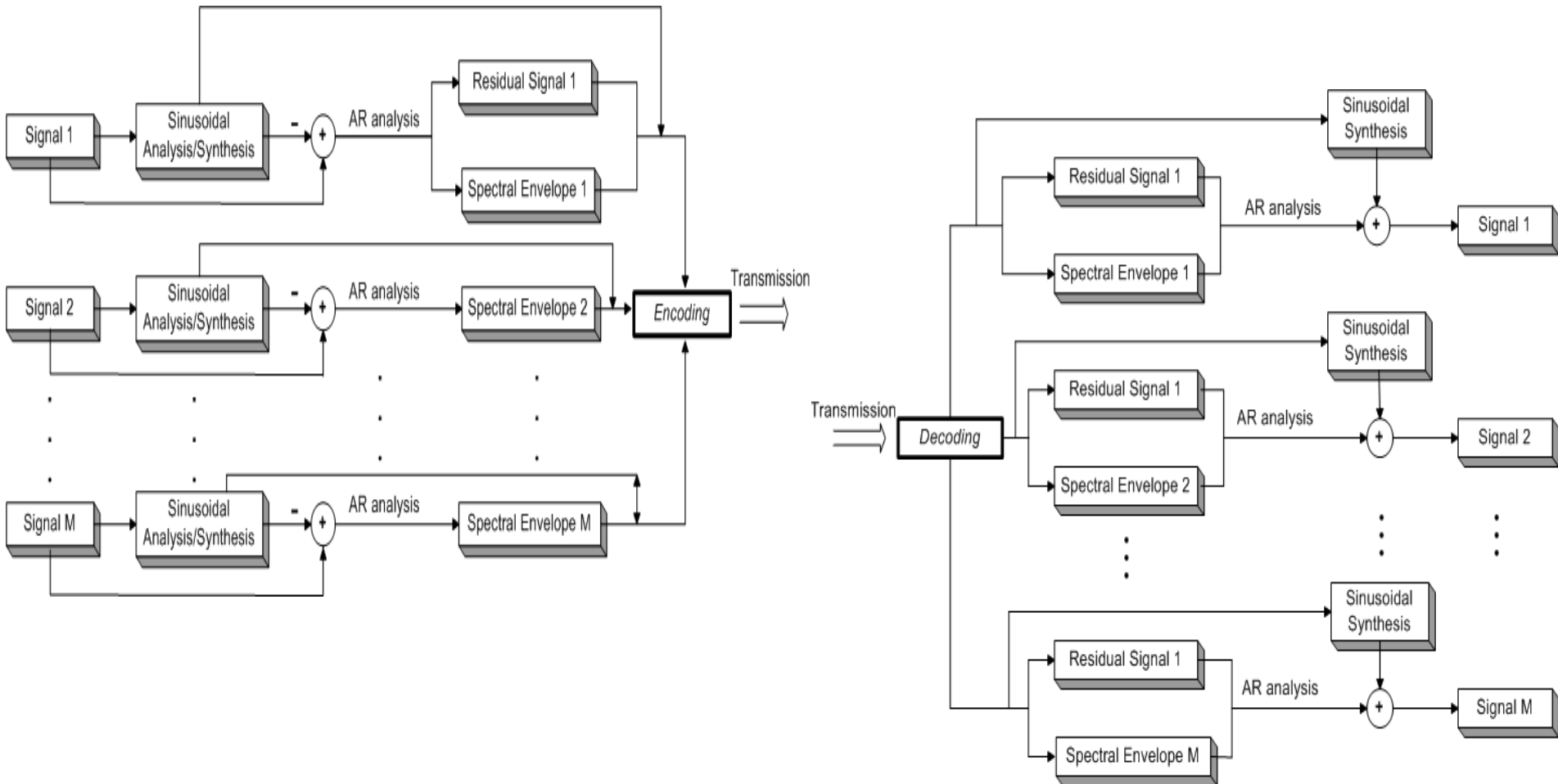
Noise transplantation (1/2)

- Consider two spot signals of a music performance
 - left channel (reference): $x_L(n)$
 - right channel (side): $x_R(n)$
- $x_L(n)$ and $x_R(n)$ have similar content
- The idea is to encode as a full audio channel the reference channel
 - side signal is reconstructed using sinusoidal parameters and LP coefficients



Noise transplantation (2/2)

- In general, consider a recording with M microphone signals



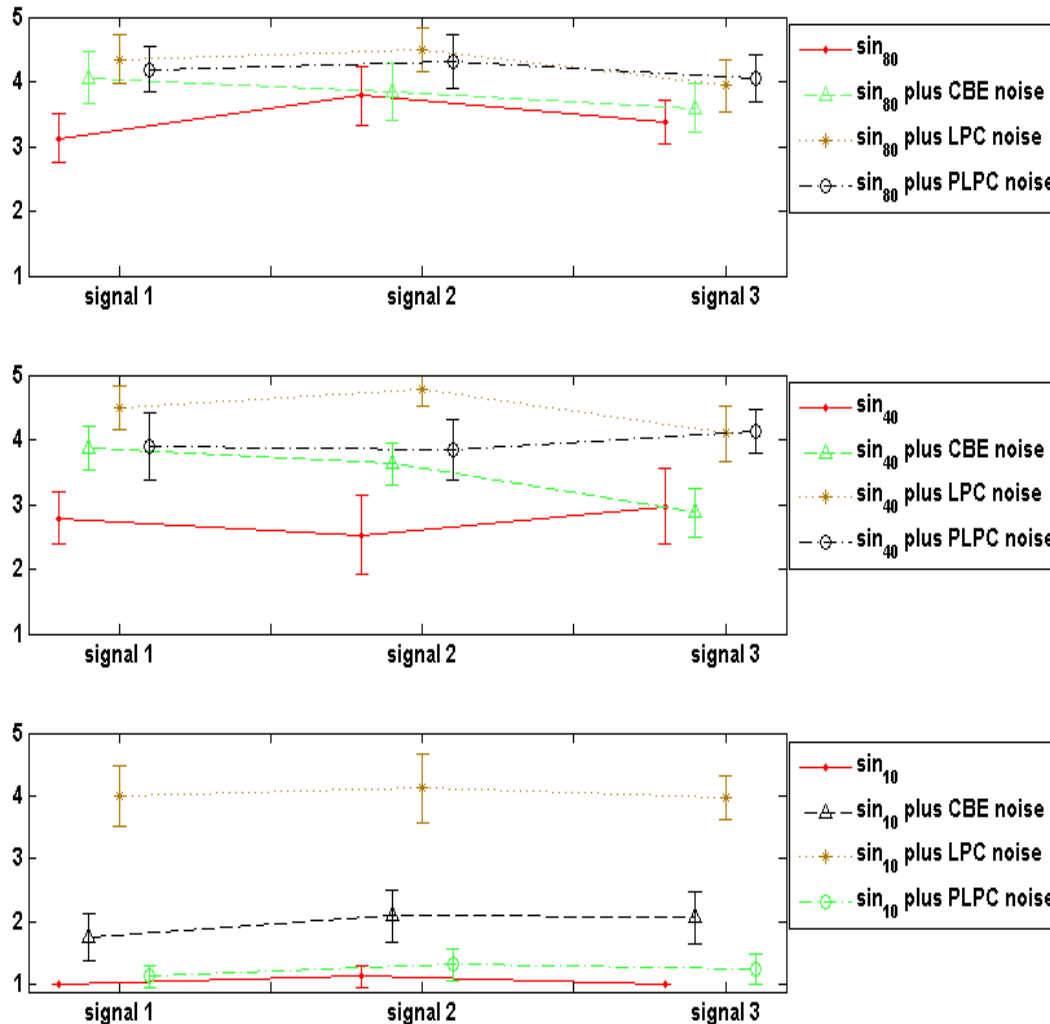
Experimental Results (1/2)

- Test with two microphone signals of a multichannel recording of a concert hall performance
 - *side channel* captures mainly the male voices of the orchestra's chorus
 - *reference channel* mainly captures the female voices
- Experimental parameters:
 - 20 msec analysis/synthesis frame
 - 50% overlapping (with overlap-add synthesis)
 - LP order for the noise shaping filters is 25
- DCR-based listening tests to compare the quality of the side signal vs. original
 - 16 listeners participated (authors are not included)
 - used three 10 sec parts from the two performances (signals 1-3)

Experimental Results (1/2)

- Noise transplantation method is compared with:
 - ***Critical Band Energy (CBE)*** approach
 - ***Perceptual Linear Predictive Coding (PLPC)*** noise shaping model
- CBE method deals with a noise perception model based on the critical bands of the auditory system
- In PLPC approach the residual is modeled by applying LPC in the perceptual domain

Experimental Results (2/2)



- The proposed method achieves high quality even for 10 sinusoids
- PLPC and CBE treat the envelope of the noise only
- The proposed method provides a residual signal for the noise part (based on similarities among spot signals)

Conclusions

- A sinusoidal plus noise model was presented tailored for modeling spot signals
- Take advantage of interchannel similarities following SAC philosophy
- Allows for remote applications based on the availability of the spot signals at the decoder side
- The perceived quality does not change by decreasing the number of the sinusoidal parameters
 - low bitrate
- Future directions:
 - examine the bitrates that are possible to achieve
 - improve transient modeling using multiresolution analysis